

Enhancing Relational Understanding in CLIP Leveraging HNC

Wen Wen

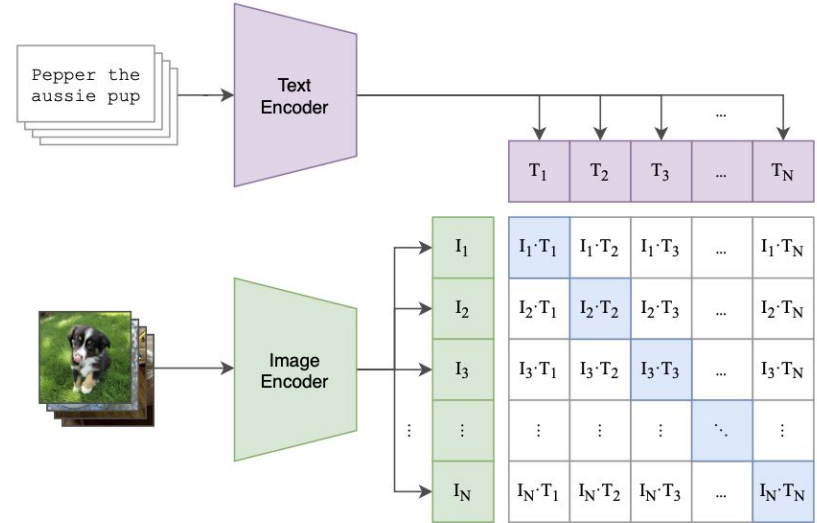
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- **Basic Knowledge**
- **Research Question**
- **Method**
- **Experiments**
- **Results and Analysis**
- **Visual Interpretability**
- **Conclusions**

Basic Knowledge

- **CLIP** (Contrastive Language–Image Pretraining), is a **dual-encoder model**
- Connect images and text in a shared embedding space
- Using a **contrastive loss**
- **Global** embedding alignment



Research Question

CILP cannot handle relations in images well.

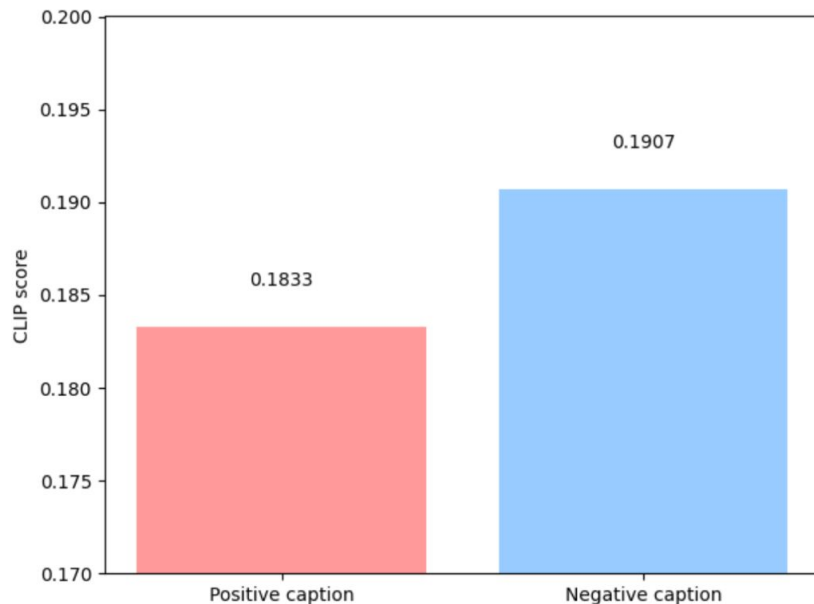
Goal: Improve relational sensitivity while preserving object recognition capabilities.



The trees are **behind** the fence

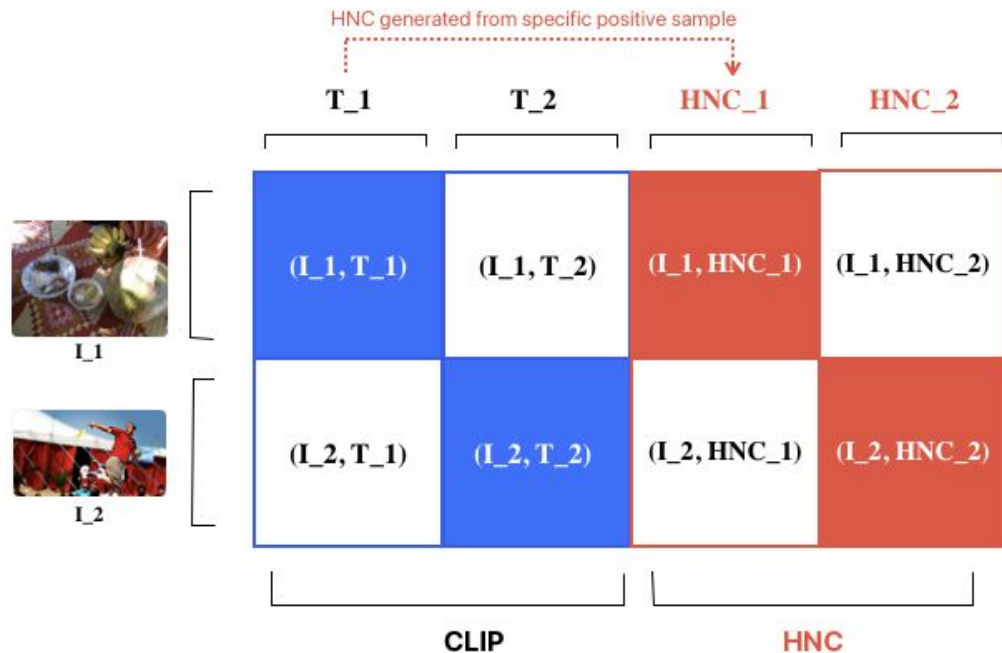


The trees are **in front of** the fence



Method:

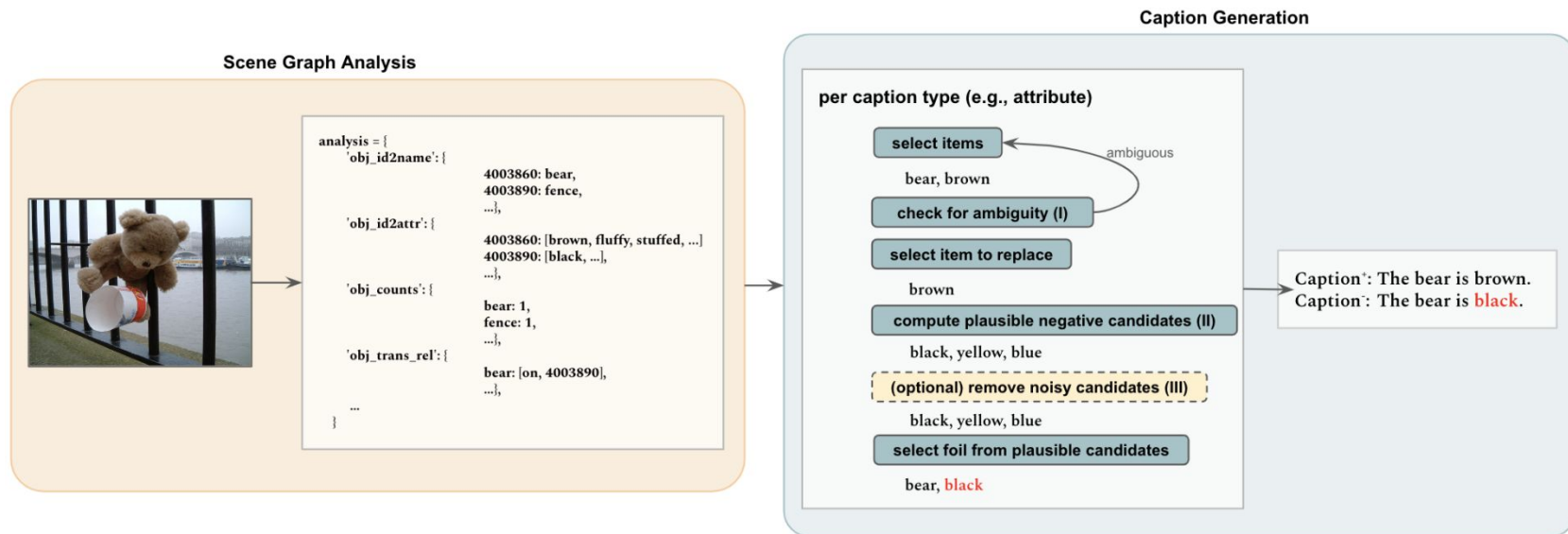
→ Inject local information for global embedding alignment
By combine contrastive learning with HNC



Blue part:
Contrastive learning for CLIP

Red Part:
Additional HNC part

Hard Negative Caption(HNC) Dataset



For each scene graph, this pipeline is run through to generate hard negative captions.

Loss Function

$$\mathcal{L}_i = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp\left(\frac{\text{sim}(l_i, T_i)}{\tau}\right)}{\exp\left(\frac{\text{sim}(l_i, T_i)}{\tau}\right) + \sum_{j \neq i} \exp\left(\frac{\text{sim}(l_i, T_j)}{\tau}\right) + \alpha \exp\left(\frac{\text{sim}(l_i, T_i^{\text{HNC}})}{\tau}\right) + \sum_{j \neq i} \exp\left(\frac{\text{sim}(l_i, T_j^{\text{HNC}})}{\tau}\right)}$$

$$\mathcal{L}_t = -\frac{1}{N} \sum_{i=1}^N \log \frac{\exp\left(\frac{\text{sim}(T_i, l_i)}{\tau}\right)}{\exp\left(\frac{\text{sim}(T_i, l_i)}{\tau}\right) + \sum_{j \neq i} \exp\left(\frac{\text{sim}(T_i, l_j)}{\tau}\right)}$$

Prevent Catastrophe Forgotten:

L2 regularization

vs

KL divergence

$$\text{Reg}_{L2} = \sum_i (\theta_i - \theta_i^{(CLIP)})^2$$

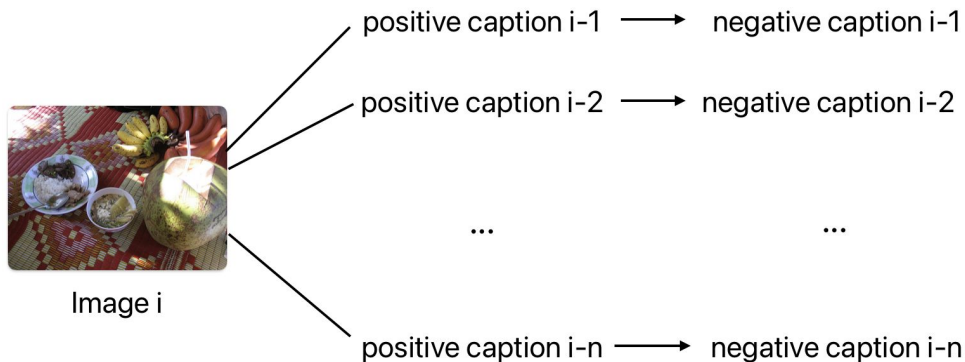
$$s_j = \frac{\exp(s_j/T)}{\sum_k \exp(s_k/T)}, \quad t_j = \frac{\exp(t_j/T)}{\sum_k \exp(t_k/T)},$$

$$D_{\text{KL}}(t \parallel s) = \sum_{j=1}^{2B} t_j \log \frac{t_j}{s_j},$$

Experiments: Datasets

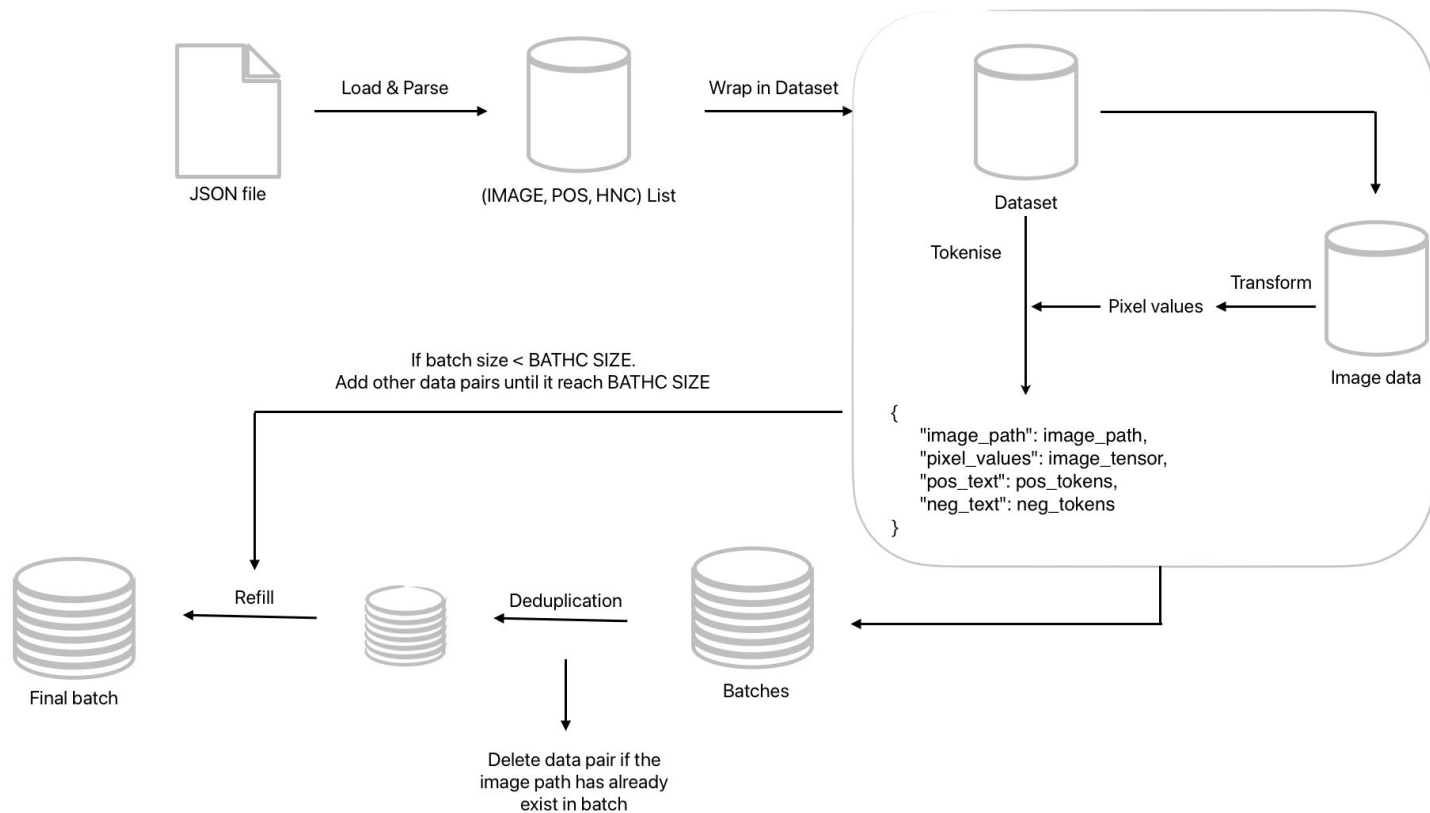
- Training: 100k samples from HNC training dataset.
- Validation: 9k samples from HNC validation dataset.
- Testing:
 - HNC human-annotated set.
 - COCO-derived relational caption pairs.

Experiments: Data Preprocessing



- One image has multiple positive captions and relative hard negative captions.
- For contrastive learning, it's important to remove the repeated image path in each batch.
- Otherwise, repeated positive captions can be treated as random negative captions for that image to influence training

Data Preprocessing



Test Dataset

- Human-annotated
- No hard mapping for positive and negative like in Train and Val DataSet

EXAMPLE:

"n6309.jpg":

```
{  
  "0": {"caption": "The people are behind the street", "type": "relation", "label": 0, "replaced": "street, car"},  
  "1": {"caption": "The people are behind the car", "type": "relation", "label": 1, "replaced": null},  
  "2": {"caption": "The horses are behind the trees", "type": "relation", "label": 0, "replaced": "behind, in front of"},  
  "3": {"caption": "The horses are in front of the trees", "type": "relation", "label": 1, "replaced": null},  
  "4": {"caption": "The car is riding the horses.", "type": "relation", "label": 0, "replaced": "men, car"},  
  "5": {"caption": "The men are riding the horses.", "type": "relation", "label": 1, "replaced": null}  
}
```

Test dataset

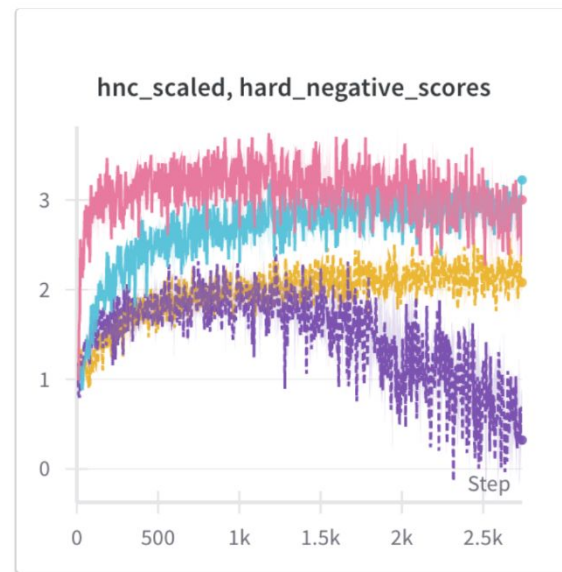
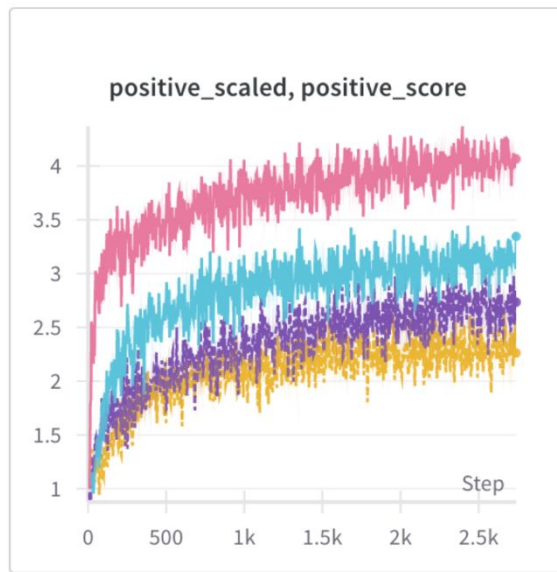
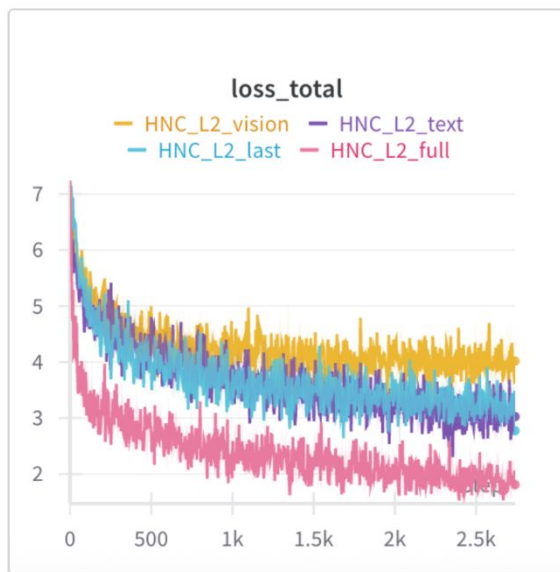
- Problem: Not always following $\{0, 1, 0, 1, 0, 1, \dots\}$
 - Grouped into consecutive pairs (i and i+1)
 - If (0, 0) or (1, 1) then skip this pair
 - If 'type' is different, then skip this pair
 - Select label=1 as positive in each pair
 - Humanly check

Test dataset: 950 samples-->After filtering: **872 samples**

Training parameters

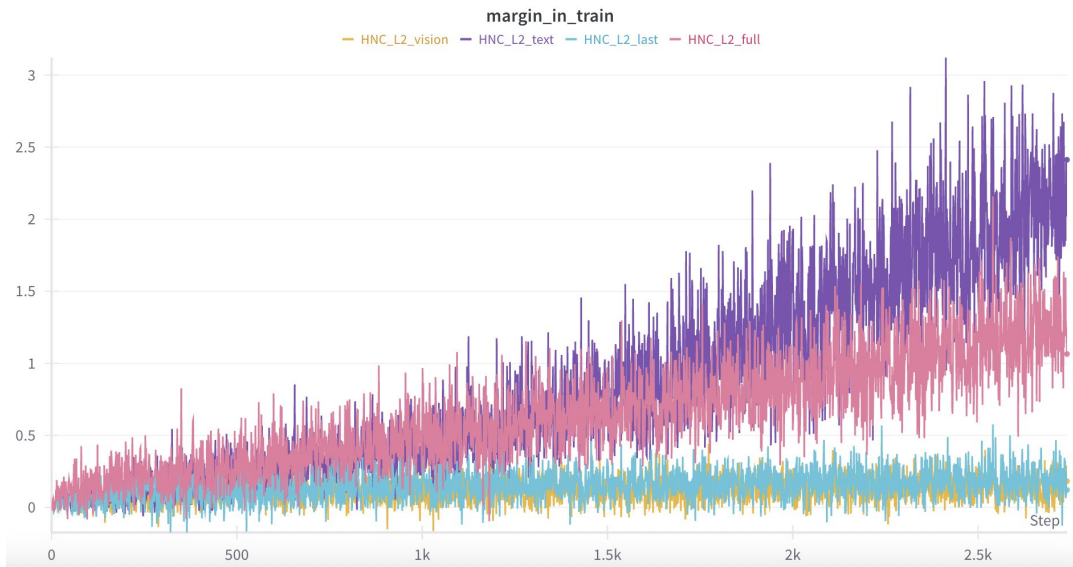
Mode	Text Encoder	Text Projection	Visual Encoder	Vision Projection
text_encoder	✓	✓	—	—
vision_encoder	—	—	✓	✓
full_encoder	✓	✓	✓	✓
last_encoder	last block	—	last block	—

Training curves across different training parameter modes.



- 20 epoch

Margin growth during training across fine-tuning modes.

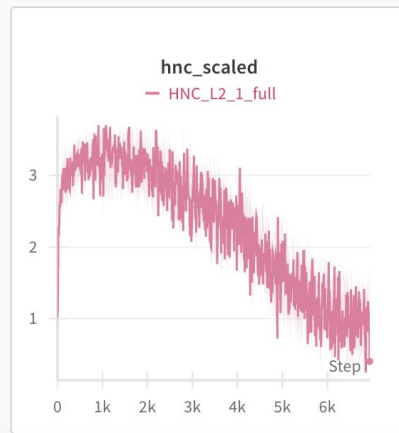
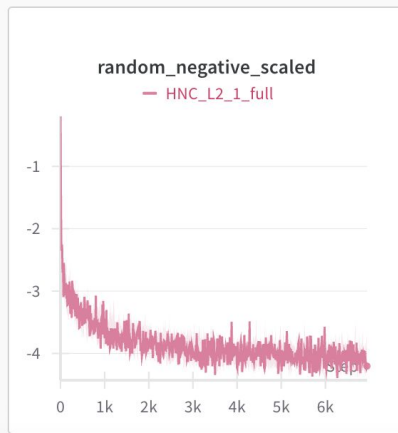
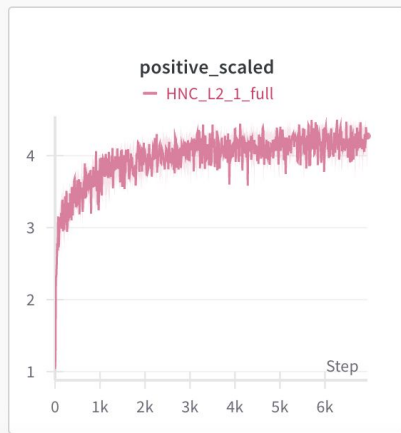
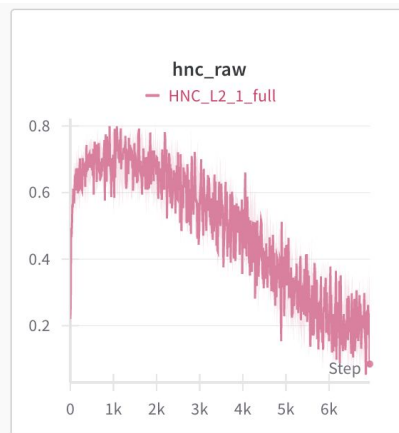
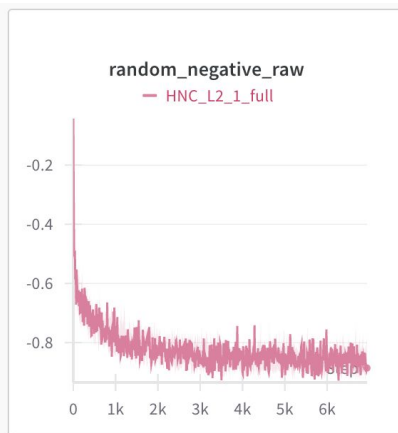
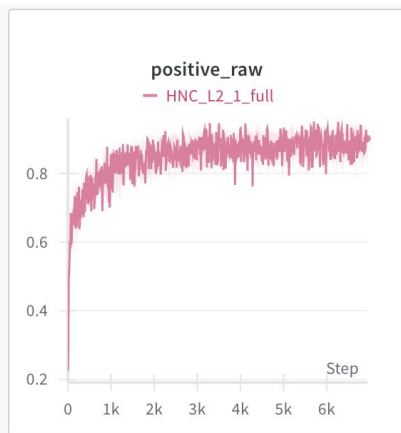


- Margin doesn't get improved during training when only train vision encoder or last encoder

Scores in Human Annotation test dataset

Model	Avg_Pos	Avg_Neg	Margin	Rand_Neg	Acc@1
ViT-B/32 (Base)	0.2478	0.2457	0.0021	0.1769	0.5459
HNC_L2_text	0.2448	0.2396	0.0052	-0.0020	0.4587
HNC_L2_vision	0.3552	0.3465	0.0087	0.0035	0.5103
HNC_L2_last	0.4639	0.4515	0.0124	0.0277	0.5264
HNC_L2_full	0.5799	0.5636	0.0163	-0.0028	0.5287

Acc@1 : positive similarity score is greater than or equal to the negative similarity score



During training:

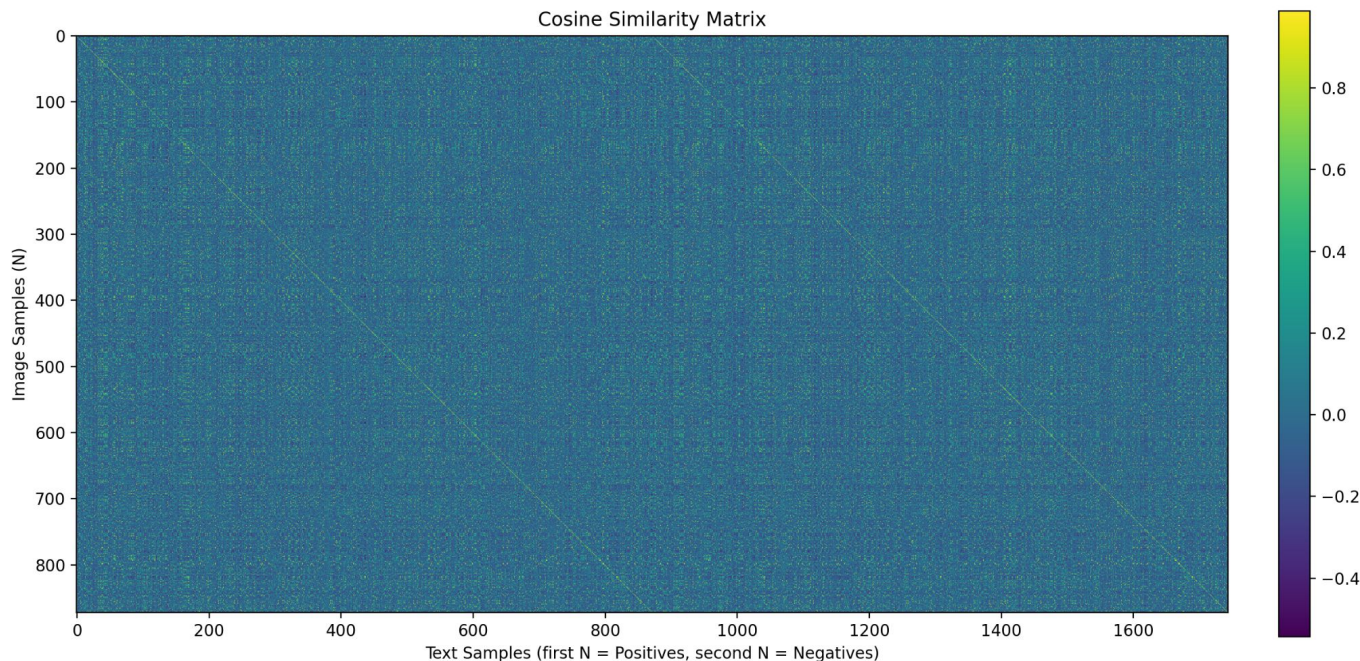
Positive score -> larger

Random negative score ->
smaller

Hard negative score -> larger
then smaller

To observe the trend, here running **50 epoch**, 137 step/epoch, $137 \times 50 = 6850$ steps

The test dataset score for fine-tuned model



Because the hard negative captions still contain a lot of image information.
So HNC scores are higher than random negative scores.

L2 regularization vs KL divergence

$$\text{Reg}_{L2} = \sum_i (\theta_i - \theta_i^{(CLIP)})^2$$

$$s_j = \frac{\exp(s_j/T)}{\sum_k \exp(s_k/T)}, \quad t_j = \frac{\exp(t_j/T)}{\sum_k \exp(t_k/T)},$$
$$D_{\text{KL}}(t \parallel s) = \sum_{j=1}^{2B} t_j \log \frac{t_j}{s_j},$$

Model	Avg_Pos	Avg_Neg	Margin	Avg_Rand_Neg	Acc@1	Acc@1.1	Acc@1.2	Acc@1.5	Acc@2	Acc@3
ViT-B/32(Base)	0.2478	0.2457	0.0021	0.1769	0.5459	0.1227	0.0470	0.0000	0.0000	0.0000
HNC_L2_text	0.2448	0.2396	0.0052	-0.0020	0.4587	0.2695	0.1927	0.1261	0.0849	0.0493
HNC_L2_vision	0.3552	0.3465	0.0087	0.0035	0.5103	0.2489	0.1755	0.0860	0.0482	0.0241
HNC_L2_last	0.4639	0.4515	0.0124	0.0277	0.5264	0.2592	0.1778	0.0963	0.0550	0.0310
HNC_L2_full	0.5799	0.5636	0.0163	-0.0028	0.5287	0.2993	0.2167	0.1227	0.0711	0.0333
HNC_KL_full	0.6023	0.5779	0.0244	0.0093	0.5631	0.2672	0.1927	0.1055	0.0608	0.0252

Performance using Coco test dataset

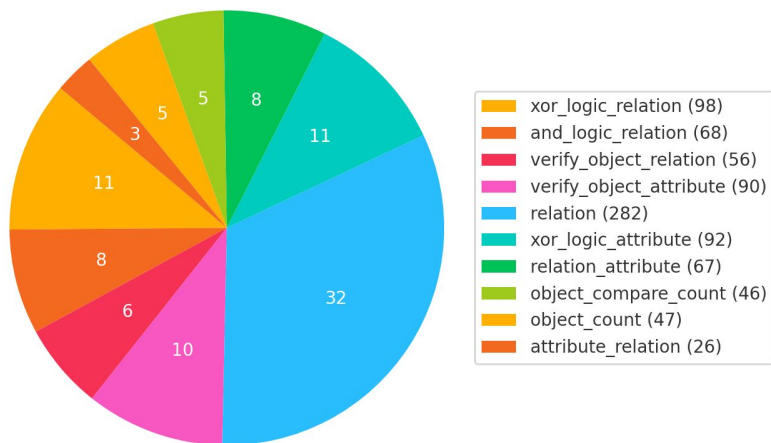
Model	Avg_Pos	Avg_Neg	Margin	Avg_Rand_Neg	Acc@1	Acc@1.1	Acc@1.2	Acc@1.5	Acc@2	Acc@3
ViT-B/32(Base)	0.3043	0.2973	0.0071	0.1510	0.7160	0.0750	0.0090	0.0000	0.0000	0.0000
HNC_L2_full	0.7198	0.7020	0.0178	0.0265	0.6180	0.1740	0.0930	0.0320	0.0150	0.0060
HNC_KL_full	0.7407	0.7255	0.0151	0.0231	0.6290	0.1170	0.0630	0.0160	0.0080	0.0000

COCO test dataset: random select 1000 samples

- Average positive score increased
- Margin increased
- Acc@1 decreased

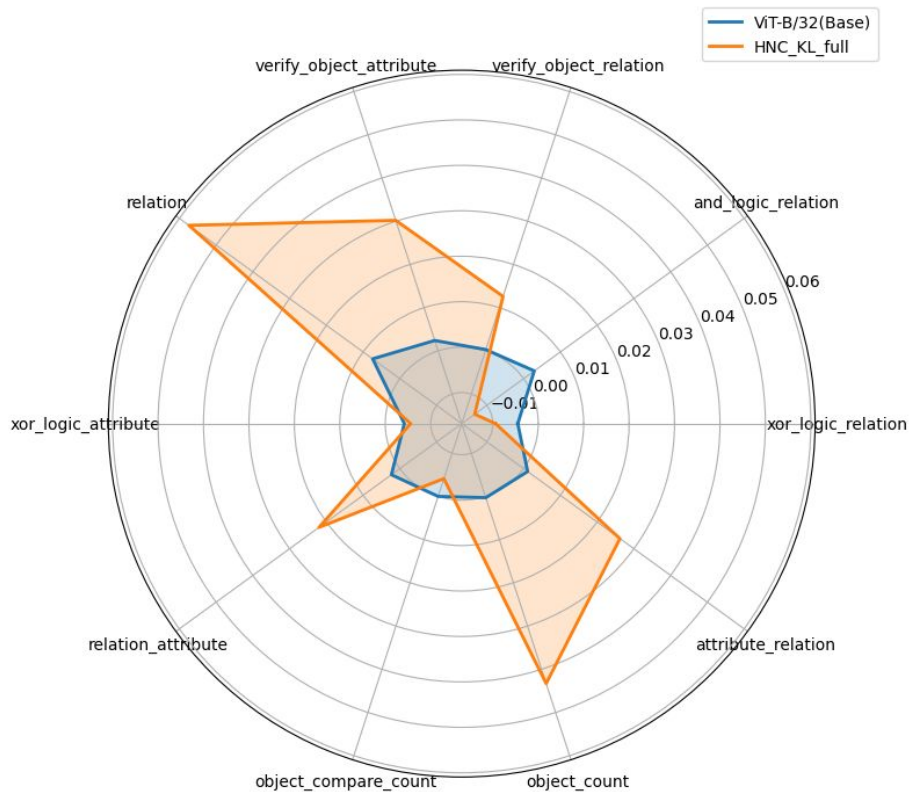
Limitations

Still struggles with complex logic and coreference.



Type	Samples	ViT-B/32(Base)	HNC_KL_full	Improvement
ALL	872	0.0021	0.0244	0.0223
relation	282	0.0074	0.0574	0.0500
xor_logic_relation	98	-0.0045	-0.0094	-0.0049
xor_logic_attribute	92	-0.0042	-0.0056	-0.0014
verify_object_attribute	90	0.0024	0.0302	0.0278
and_logic_relation	68	0.0029	-0.0133	-0.0162
relation_attribute	67	0.0023	0.0220	0.0197
verify_object_relation	56	0.0003	0.0126	0.0123
object_count	47	0.0003	0.0433	0.0430
object_compare_count	46	0.0000	-0.0041	-0.0041
attribute_relation	26	0.0011	0.0262	0.0251

Each 'type' score



Bad at:

- 'and_logic_relation'
- 'xor_logic_relation'
- 'object_compare_count'
- 'xor_logic_attribute'

Examples of Challenging Type

Type - **and_logic_relation**

Pos: There are both a lamp behind the women and a laptop in front of the window.

Neg: There are both a lamp in front of the women and a laptop in front of the window.

Type - **xor_logic_relation**

Pos: The laptop is either behind the women or the sofa.

Neg: The laptop is either behind the women or the foot.

Type - **object_compare_count**

Pos: There are as many remote controls as people.

Neg: There are fewer remote controls than people.

Type - **xor_logic_attribute**

Pos: There is either a red carpet or green curtains.

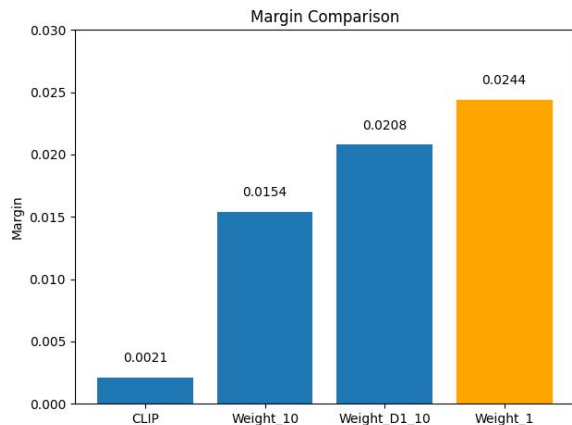
Neg: There is either a red carpet or purple curtains.



Limitations

Overweighting hard negatives may hurt stability.

- Using larger HNC weight doesn't performance well.
- Using dynamic weight (small at first, increasing the weight during training), better than fixed large weight, but still performance bad.



Visual Interpretability

Based on *Explaining Caption-Image Interactions in CLIP models with Second-Order Attributions*
(Tilli et al.)

Pos: The trees are behind the fence
- text-to-image

Original Image



Base CLIP
score=0.0430



Fine-tuned CLIP
score=0.0504



<https://arxiv.org/abs/2408.14153>

Visual Interpretability

Pos: The trees are behind the fence

- image-to-text

Base CLIP

the trees are behind the fence

Fine-tuned CLIP

the trees are behind the fence

Visual Interpretability

Pos: The **trees** are behind the fence.

Base CLIP: "trees"



Fine-tuned CLIP: "trees"



Visual Interpretability

Pos: The trees are behind the **fence**.

Base CLIP: "fence"



Fine-tuned CLIP: "fence"



→ Object detection get improved. E.g. trees, fence

Visual Interpretability

Pos: The trees are **behind** the fence.

Base CLIP: "behind"



Fine-tuned CLIP: "behind"



→ For relational words (e.g. behind) can also focus on the right place in image.

Results Summary

- Relational understanding improved
- Text encoder plays a central role
- Full encoder tuning achieves the best overall performance
- KL regularization yields highest margin; L2 excels at strict thresholds.
- Good out-of-domain generalization (On a COCO-derived test set)
- Visual interpretability improved

Thank you